Wavetable Synthesis

Intro to Audio Tech 1
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Monday, September 15

Georgia Tech

Lesson Outline

- Checkpoint 3
- Wavetable Synthesis
- Wavetable Synthesis in Max
- Vector Synthesis
- Up Next…

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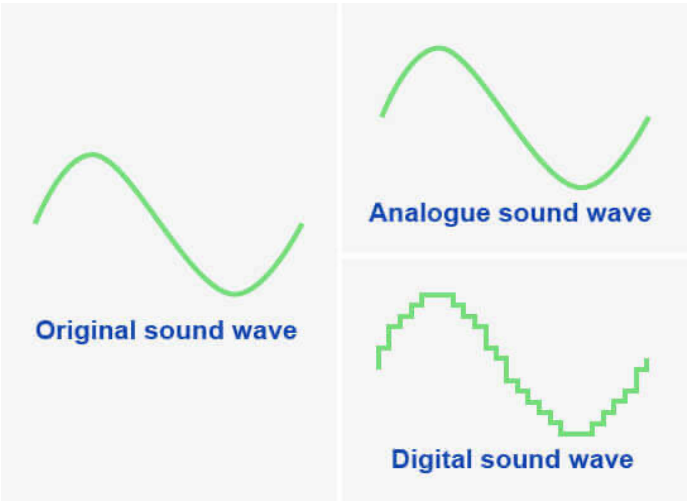
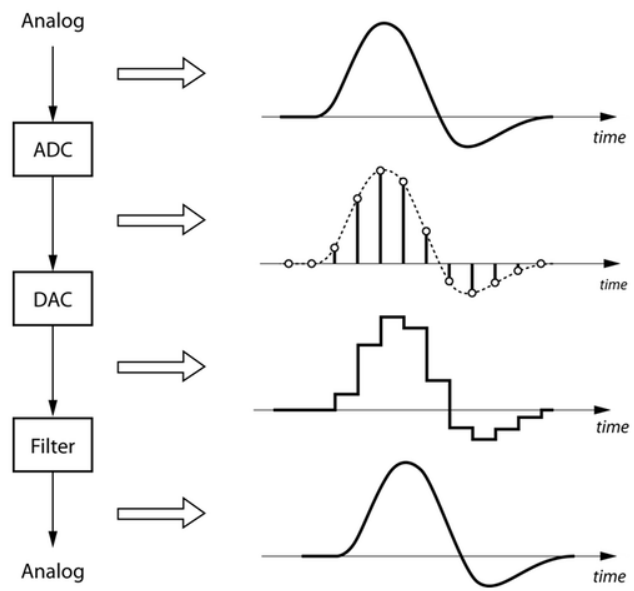
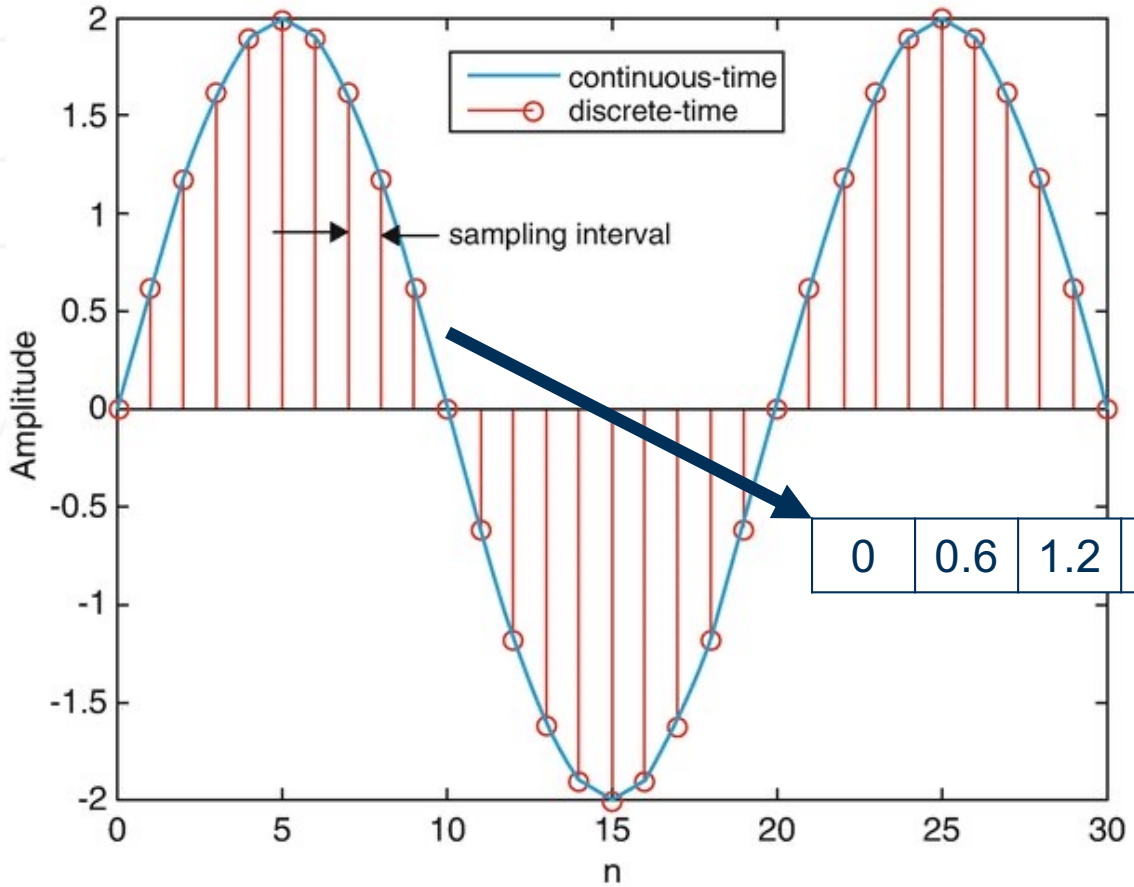
Digital Audio

- Continuous vs Discrete
- A digital audio signal is just a sequence of numbers





Audio as Samples



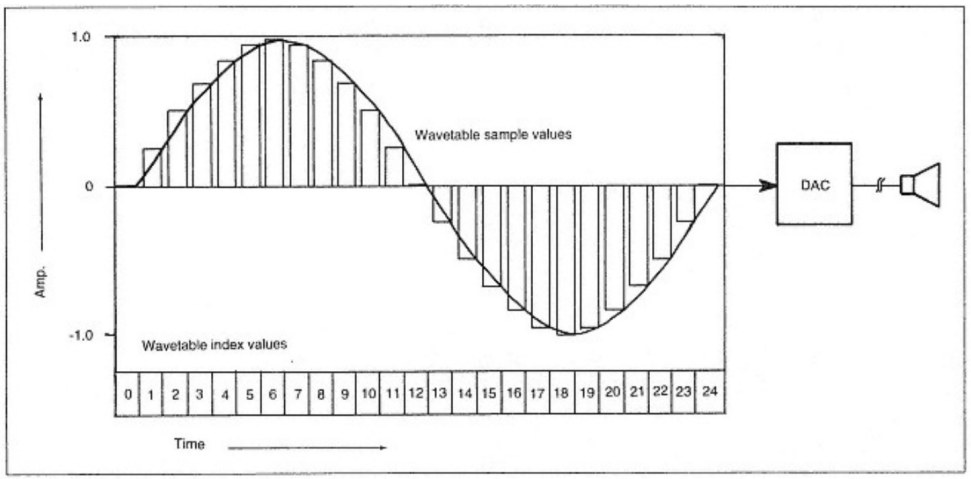
0	0.6	1.2	1.6	1.9	2	1.9	1.6	1.2	0.6	0

$x[n] = \sin(n)$

Wavetable Synthesis

Wavetable Synthesis

- Store waveform (full cycle) in a buffer/table/array
- Function to read values from the wavetable
- We keep reading the values from start to end to start to create a ”continuous” waveform

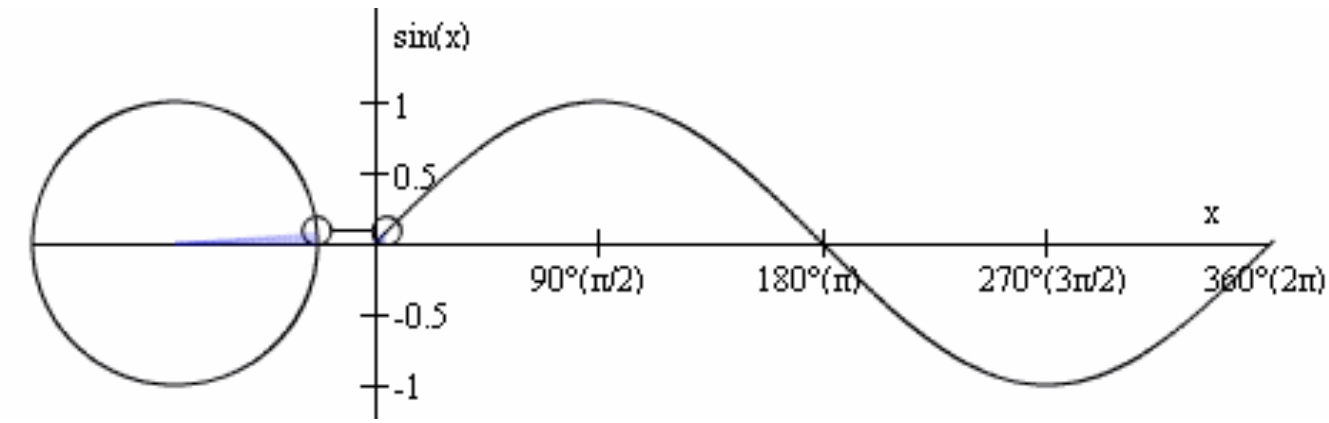


Why Wavetable Synthesis

- Less computation time per sample
  - Avoid calculating a sinusoid 48k times a second
- Flexibility for waveform storage
- Synthesis possibilities!
  - Combine waveforms, enveloping, looping, sampling synthesis, etc

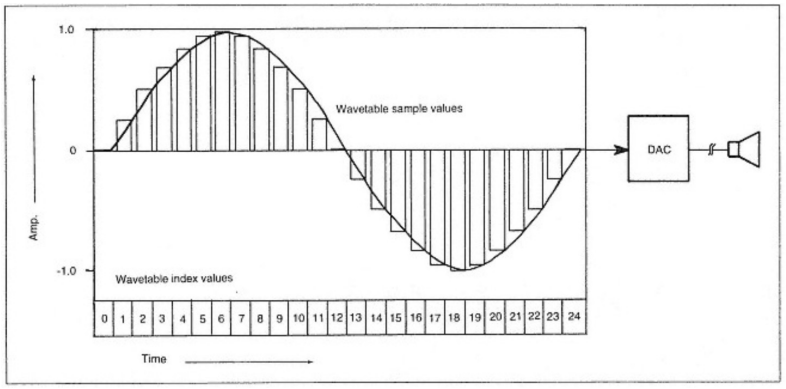
How it works

- Store waveform (full cycle) in a buffer/table
- Function to read values from the wavetable
  - The read point location is analogous to phase in waveform
- How do we change the waveform frequency?



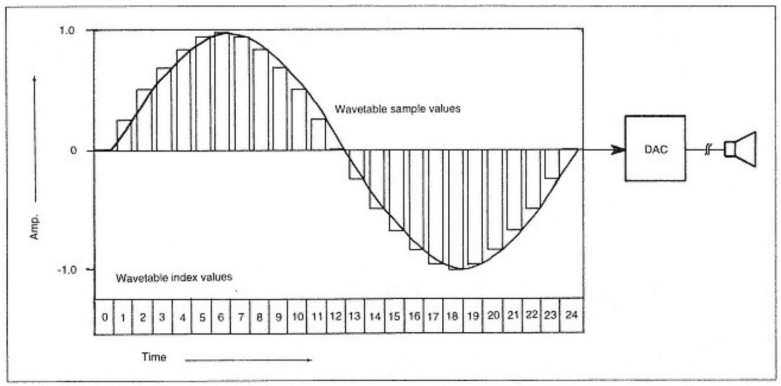
Wavetable Synthesis: Changing Frequency

- If we readthrough a wavetable one time in a second, what's the output frequency
  - 1 Hz
- If we read it 100 times in a second, we'd get…
  - 100 Hz
- However, this gets more complicated as we consider wavetable length and the sampling frequency



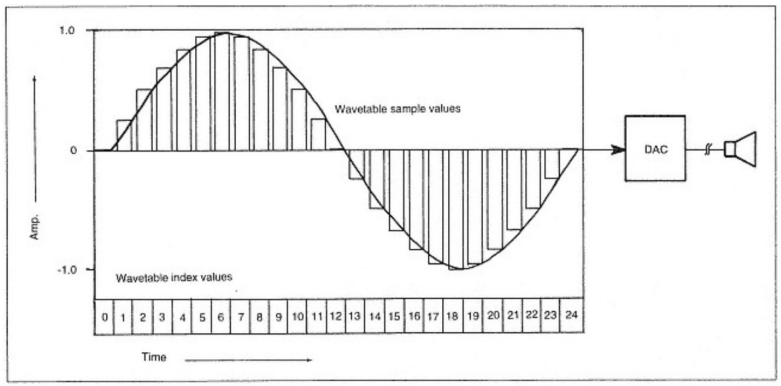
Wavetable Synthesis: Changing Frequency

- If we read through our 25-length wavetable at 25,000 samples/s, what is the frequency?
  - 25,000/25= 1,000 Hz
- If we change the sampling frequency to 50,000 samples/s, we get…
  - 50,000/25 = 2,000 Hz
- So, we can just change the read frequency (or sampling frequency)



Wavetable Synthesis: Changing Frequency

- But changing the sampling rate of your system isn't always practical and can cause trouble
- Instead, we can:
  - Skip samples in the table (pitch up)
  - Repeat samples in the table (pitch down)
  - Interpolate new samples
- This "stretches" the wavetable size to generate new frequencies



Basic Wavetable ”Resampling” Algorithm

Read_index = (prev_index + increment) % table_length

Output = amplitude_control * wavetable[read_index]

- The frequency of playback is dependent on the table length, sampling rate, and increment

$$inc = \frac{table_length * frequency}{sampling_rate}$$

$$frequency = \frac{sampling_rate * inc}{table_length}$$

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Wavetable Example

- If sampling rate is 40,000 samples/s, table length is 1000 samples, and we want an output frequency of 2000 Hz, then…

$$inc = \frac{table_length * frequency}{sampling_rate} = \frac{1000 * 2000}{40{,}000} = 50\ samples$$

- Our increment will be 50 samples

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Wavetable Interpolation Example

- Our numbers may not always work out into whole increments
- If sampling rate is 40,000 samples/s, table length is 25 samples, and we want an output frequency of 2000 Hz, then…

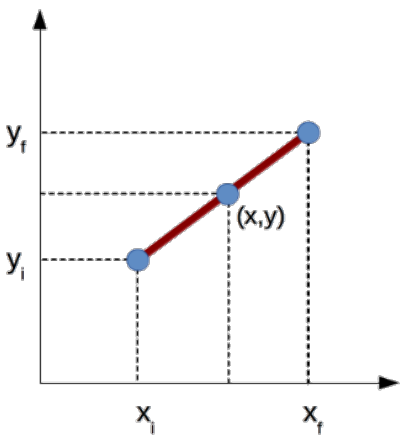
$$inc = \frac{table_length * frequency}{sampling_rate} = \frac{25 * 2000}{40{,}000} = 1.25\ samples$$

- In this case, we could truncate/round our read position which will cause some noise
- Or we can linearly interpolate between the known indexes and the increment

Wavetable Interpolation Example

$$inc = \frac{table_length * frequency}{sampling_rate} = \frac{10 * 2000}{40{,}000} = 0.5\ samples$$

Ind	0	1	2	3	4	5	6	7	8	9	10
Val	0	0.6	1.2	1.6	1.9	2	1.9	1.6	1.2	0.6	0



$$val_0 + \frac{(inc - ind_0)(val_1 - val_0)}{ind_1 - ind_0}$$

$$0 + \frac{(0.5 - 0)(0.6 - 0)}{1 - 0} = 0.3$$

Increment	Output Val
0	0
0.5	0.3
1	0.6

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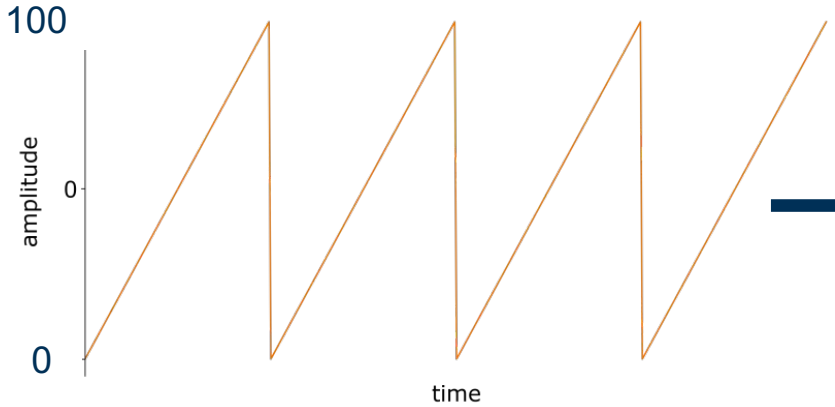
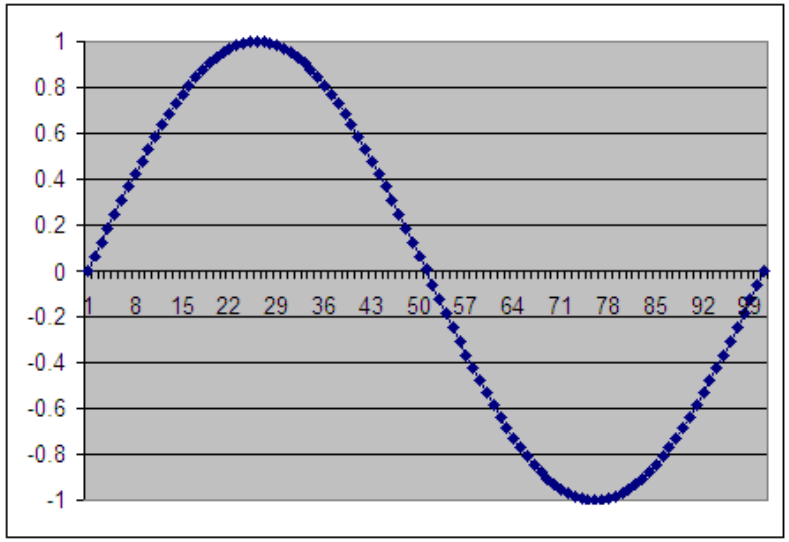
Standard Wavetable Lookup

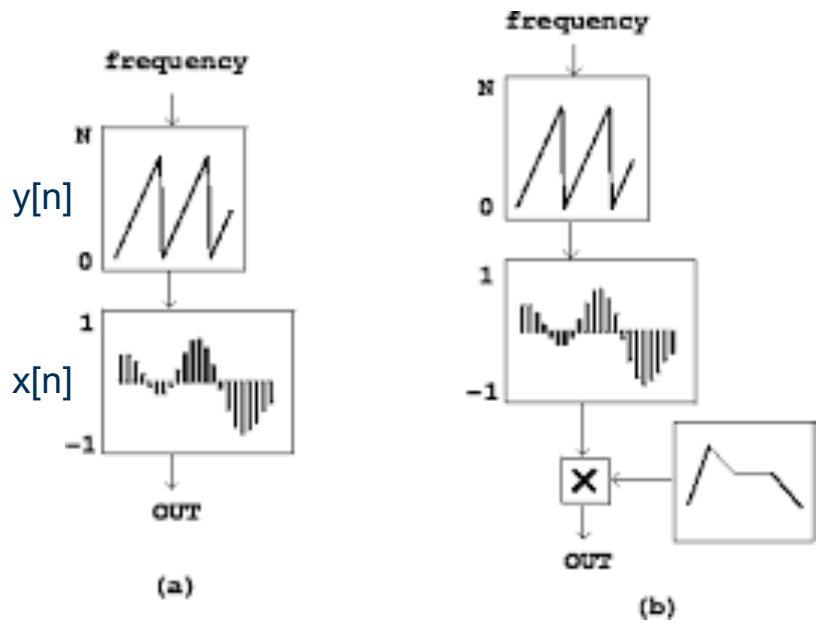
- Phasor
  - produces a repeated ramp from 0-N at some frequency
  - Sawtooth wave or ramp function



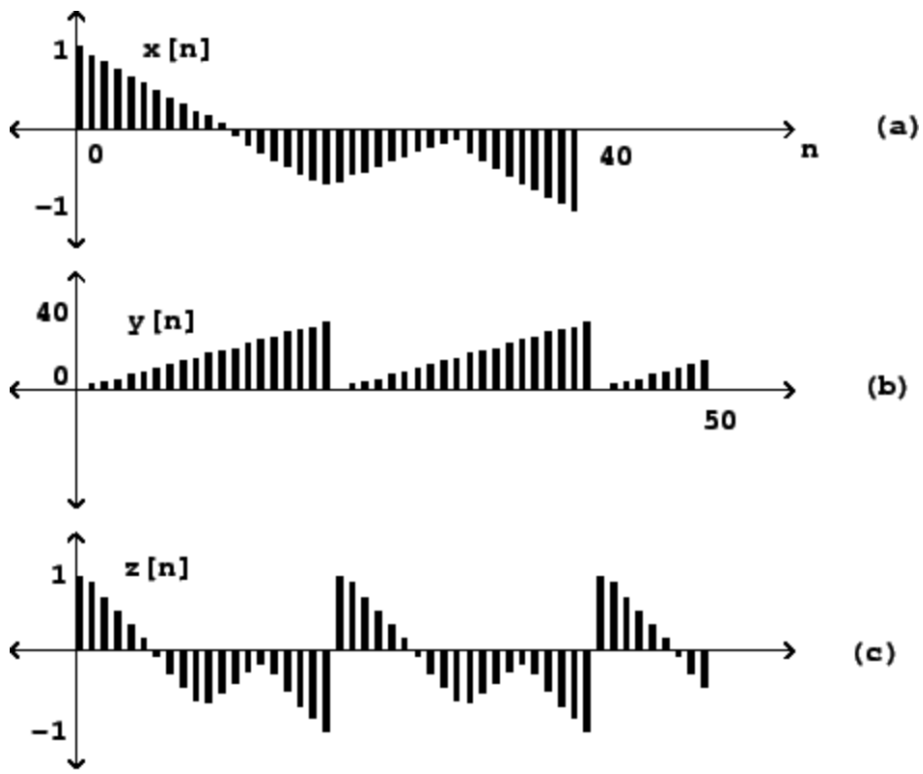


Wavetable Synthesis

- Stored digital waveform that can be read through and looped repeatedly
- Look-up table
- z[n] = x[y[n]]



Wavetable Lookup



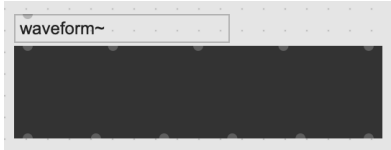
- Signal b (y[n]) is being used to read through signal a (x[n])
- This gives signal c (z[n])
- Notice that signal a has range -1 to 1 with length 40 and signal b has range 0-40

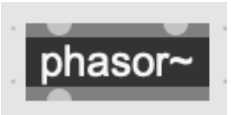
Z[n] = x[y[n]]

Wavetable In Max

Wavetable Synthesis

- We can store samples in a managed memory array called a buffer~
- We can view buffer~ contents in the waveform~ object
- In Max, we don’t need to do all the hard work of calculating read frequency, increment indexing, and interpolation
- Instead, we use the wave~ object which specializes in reading wavetables along with phasor~

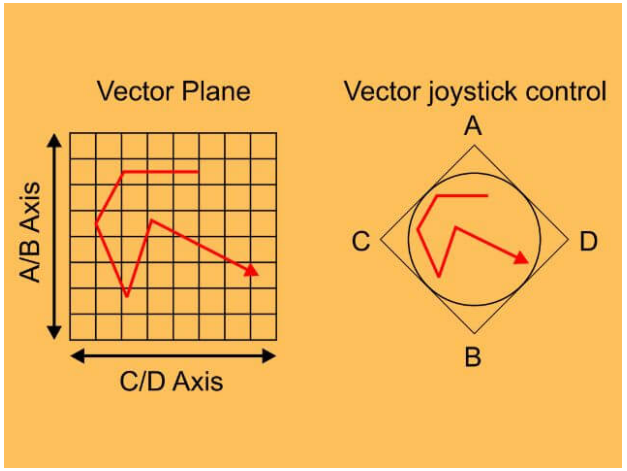




Vector Synthesis

Vector Synthesis

- Reads between multiple wavetables to create more complex timbres
- Also called wavetable crossfading and linear algorithmic synthesis
- Implement by controlling amplitude presence and crossfade time between wavetables
- Typically done with 4 wavetables
- Lays the path for other wavetable synthesis techniques like
  - Wave Stacking
  - Wave Terrain Synthesis



Up Next…

- Study Guide is out for Exam 1
- Assignment 2 is due next Friday at 11:59pm
- Exam next Wednesday